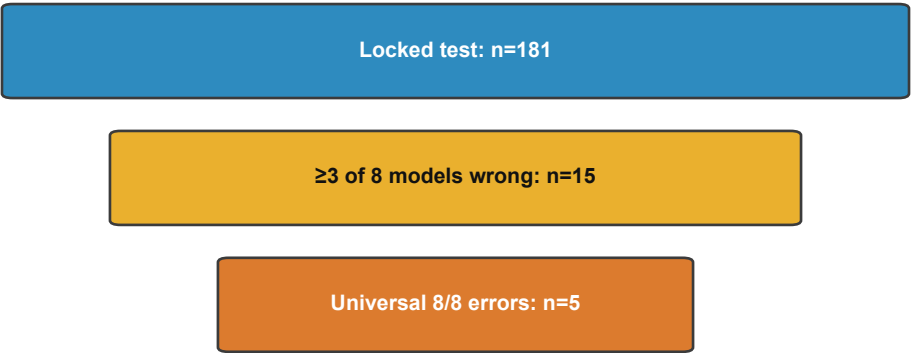


Figure 5. Anatomy of universal model failures

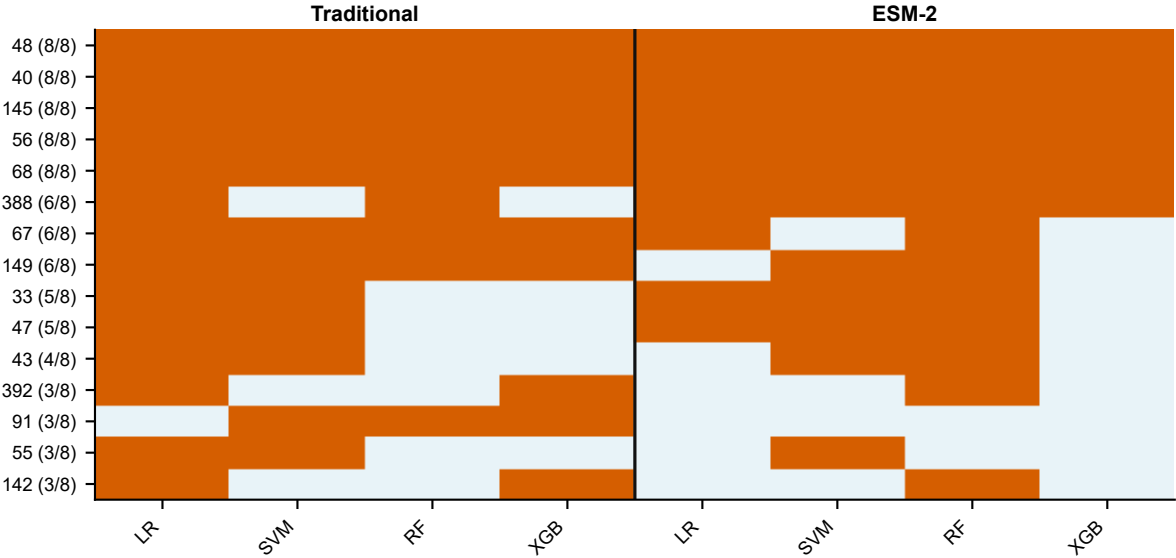
A

Hard-case hierarchy



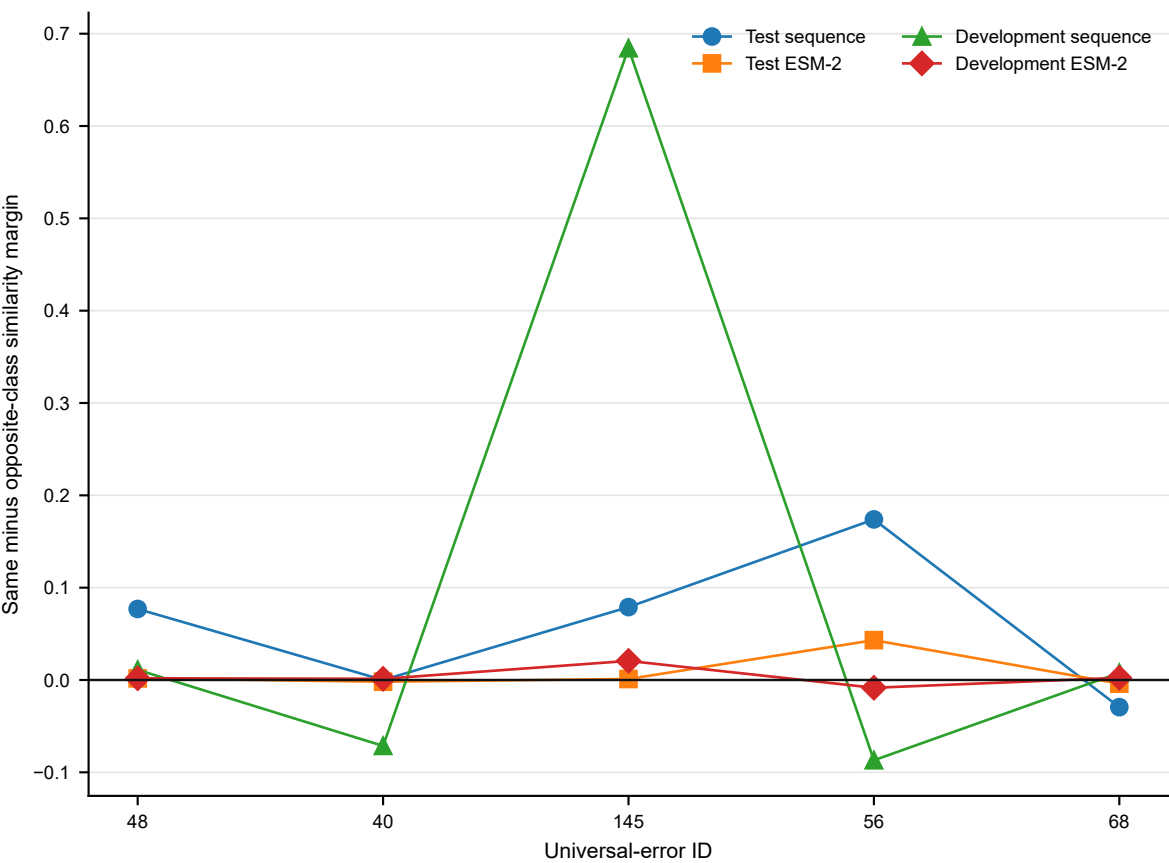
B

Eight-model error patterns among 15 hard cases



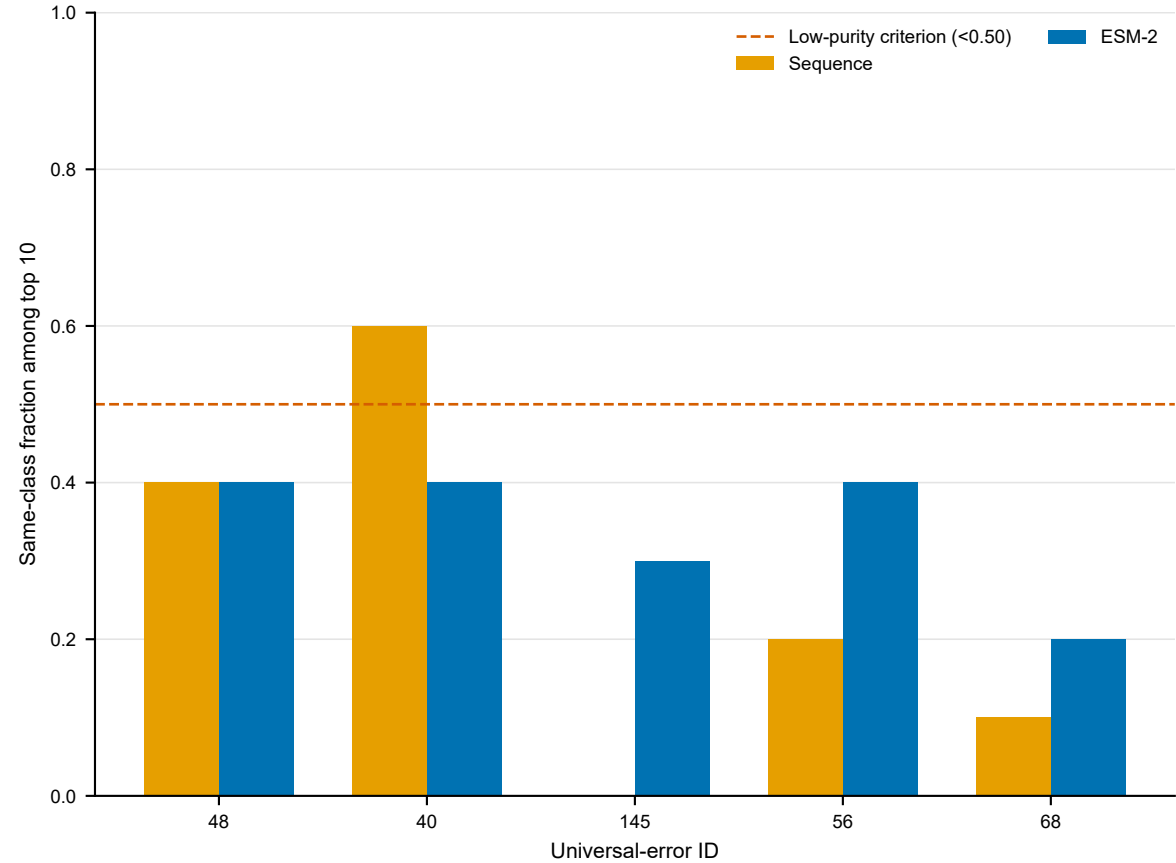
C

Neighborhood conflict margins



D

Development-neighborhood purity



E

Integrated universal-error evidence map



F

Universal-error sequence and residue summary

ID	Class	Top residue	Sensitivity	Motif context	Evidence category
48	Inactive	L9	0.832	LK;KAL	Cross-class neighborhood conflict
40	Inactive	F12	0.742	FA	Mixed/ambiguous neighborhood
145	Inactive	K8	0.487	KK	Cross-class neighborhood conflict
56	Inactive	F1	0.233	FA	Same-class neighborhood but model-sensitive
68	Active	K14	0.187	AK;LAK	Cross-class neighborhood conflict

Descriptive model-interpretability evidence; not evidence of label error or biochemical causality.